Positive powers are often easier to use, and they are *required* by some z-domain techniques. Why not just rewrite Eq. 33-3 using positive powers and forget about negative powers entirely? We can't! The trick of dividing the numerator and denominator by the highest power of z (such as z^4 in our example) can only be used if the number of recursion coefficients is already known. Equation 33-3 is written for an *arbitrary* number of coefficients. The point is, both positive and negative powers are routinely used in DSP and you need to know how to convert between the two forms.

The transfer function of a recursive system is useful because it can be manipulated in ways that the recursion coefficients cannot. This includes such tasks as: combining cascade and parallel stages into a single system, designing filters by specifying the pole and zero locations, converting analog filters into digital, etc. These operations are carried out by algebra performed in the s-domain, such as: multiplication, addition, and factoring. After these operations are completed, the transfer function is placed in the form of Eq. 33-3, allowing the new recursion coefficients to be identified.

Just as with the s-domain, an important feature of the z-domain is that the transfer function can be expressed as **poles** and **zeros**. This provides the second general form of the z-domain:

EQUATION 33-4
Transfer function in pole-zero form.

$$H[z] = \frac{(z - z_1)(z - z_2)(z - z_3)\cdots}{(z - p_1)(z - p_2)(z - p_3)\cdots}$$

Each of the poles (p_1, p_2, p_3, $\cdots$) and zeros (z_1, z_2, z_3 $\cdots$) is a complex number. To move from Eq. 33-4 to 33-3, multiply out the expressions and collect like terms. While this can involve a tremendous amount of algebra, it is straightforward in principle and can easily be written into a computer routine. Moving from Eq. 33-3 to 33-4 is more difficult because it requires *factoring* of the polynomials. As discussed in Chapter 32, the quadratic equation can be used for the factoring if the transfer function is second order or less (i.e., there are no powers of z higher than z^2). Algebraic methods cannot be used to factor systems greater than second order and numerical methods must be employed. Fortunately, this is seldom needed; digital filter design *starts* with the pole-zero locations (Eq. 33-4) and *ends* with the recursion coefficients (Eq. 33-3), not the other way around.

As with all complex numbers, the pole and zero locations can be represented in either polar or rectangular form. Polar notation has the advantage of being more consistent with the natural organization of the z-plane. In comparison, rectangular form is generally preferred for mathematical work, that is, it is usually easier to manipulate: $\sigma + j\omega$, as compared with: $re^{j\omega}$.

As an example of using these equations, we will design a notch filter by the following steps: (1) specify the pole-zero placement in the z-plane, (2)